

Comments – Key Principles

Comments exist because code sometimes fails to express intent. The goal is always to write code that makes comments unnecessary. Code is the single source of truth, while comments tend to become outdated.

General Philosophy

Do not use comments to explain bad code. Rewrite the code instead. Clear naming, small functions, and proper structure reduce the need for comments.

Good Use of Comments

Used sparingly, comments can add value when they explain intent, clarify ambiguity, or warn about consequences.

Example – Explanation of Intent

```
// GOOD: explains design decision
return 1; // ensure this type is always sorted higher
```

Example – Clarification

```
// GOOD: translates unclear return value
assertTrue(a.compareTo(b) == -1); // a < b
```

Example – Warning

```
// GOOD: warns about hidden risk
// SimpleDateFormat is not thread-safe
```

Example – Amplification

```
// GOOD: highlights critical detail
value = input.trim(); // required to avoid parsing error
```

Bad Comments

Most comments fall into this category. They are redundant, misleading, or simply noise.

Example – Redundant

```
// BAD: repeats obvious code
int age; // age of the person
```

Example – Noise

```
/**
 * Returns the day of the month.
 */
public int getDayOfMonth() { }
```

Example – Misleading

```
// BAD: not accurate
// returns when closed
waitForClose();
```

Example – Mandated

```
/**
 * @param title The title
 * @param author The author
 */
void addCD(String title, String author)
```

Example – Commented-out Code

```
// BAD: dead code left behind
// int oldValue = calculate();
```

Example – Nonlocal Comment

```
// BAD: refers to external system behavior
// uses default port 8080
```

Better Alternative: Replace Comments with Code

```
// BAD
// check if eligible
if ((emp.flag & HOURLY) && emp.age > 65)

// GOOD
if (emp.isEligibleForFullBenefits())
```

Key Takeaways

Write code that explains itself. Use comments only when necessary. Keep them accurate, local, and meaningful. Remove anything that does not add value.